

Nutrition Management in Geriatric Patients with CKD

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Sarcopenia, defined as age-related decline in skeletal muscle mass and functional capacity, is a hallmark nutritional abnormality observed in patients with moderate-to-advanced CKD. Uremic state and associated medical conditions also predispose older patients with CKD to protein–energy wasting, a nutritional abnormality that could include sarcopenia. Prevention of protein and energy depletion and replenishing the already low nutritional reserves elderly patients with CKD should focus on conventional and innovative strategies. This review aims to provide an overview of the mainstay of nutritional therapy in this patient population, such as intake of adequate amounts of protein and energy along with preserving fluid, electrolyte, and mineral balance, and to discuss more innovative interventions to aid these approaches.

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Introduction

Nutritional management is one of the most important aspects of health in patients with CKD, especially in ones with moderate-to-advanced stage. The increasing risk of protein–energy wasting is well established in these vulnerable patients and stems from various factors including but not limited to inadequate dietary nutrient intake relative to increased needs, concurrent metabolic abnormalities leading to altered nutrition utilization, and the detrimental effects of kidney replacement therapies on nutritional status and metabolic milieu. CKD is also a disease of the aging population, who are independently at risk of protein–energy wasting and sarcopenia. While some of the factors that lead to these derangements overlap between CKD and aging, there are distinct elements associated with each condition, which in combination could exponentially increase the risk of poor clinical outcomes, such as cardiovascular disease, infection risk, and frailty. In this review, we will discuss epidemiology of protein–energy wasting (PEW) in the elderly patients with CKD and provide insights into the mechanisms that lead to this abnormal state. We will also discuss the strategies aimed at preventing and treating PEW, including recommendations for managing fluid and electrolyte balance in elderly patients with CKD.

Epidemiology of PEW and Sarcopenia in Elderly Patients with CKD

Aging is generally measured with chronologic age, and older age is most commonly defined as 65 years or older.¹ However, the relationship between aging and CKD becomes more intricate because of the occurrence of premature aging induced by kidney disease itself. Indeed, systemic complications seen in CKD, including

chronic inflammation, oxidative stress, and uremia, impair antiaging mechanisms and promote aging.² A significant consequence of this process is poor nutritional status and accelerated muscle wasting. Both advanced kidney disease and increasing age are associated with net protein catabolism and progressive muscle loss, which are exacerbated by inadequate nutritional intake frequently observed in elderly patients with CKD.

Sarcopenia and protein–energy wasting are two related but distinct terminologies commonly used to describe the loss of muscle mass and function seen in elderly patients with CKD. Sarcopenia is defined as age-related decline in skeletal muscle mass and functional capacity and is a hallmark nutritional characteristic of aging.^{3,4} Various different diagnostic criteria are available to define sarcopenia, including the European Working Group on Sarcopenia in Older People⁵ and International Working Group on Sarcopenia.⁶ These criteria identify sarcopenia on the basis of criteria such as low overall or appendicular skeletal muscle mass in conjunction with poor functional status. Accordingly, the estimated prevalence of sarcopenia is around 10% in the healthy population aged 60 years and older.⁷ Sarcopenia is substantially more prevalent among patients with CKD. In a cross-sectional study by Lamarca *et al.*, the prevalence of sarcopenia varied between 4% and 63% depending on the definition of sarcopenia in patients aged 60 years and older on maintenance hemodialysis while the prevalence of decreased muscle strength was found to be 85%.⁸ The methods used to define low muscle mass included appendicular lean mass index measured by dual-energy x-ray absorptiometry; lean body mass index (BMI) measure by bioelectrical impedance or skinfold thickness; mid-arm muscle circumference, calf circumference, and muscle function assessed by

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handgrip strength.⁸ The prevalence of sarcopenia defined with the European Working Group on Sarcopenia in Older People criteria using lean mass index and handgrip strength was 37% in another multicenter observational study.⁹

PEW refers to the loss of body protein and fuel reserves associated with kidney disease. Diagnosis of PEW is historically based on the presence of low serum biomarkers (e.g., serum albumin or transthyretin), decrease in body weight or muscle mass (e.g., reduced mid-arm muscle circumference), or weight loss with reduced intake of protein and energy.¹⁰ The development of PEW in CKD is driven by many contributing factors, which include but not are limited to poor nutritional intake, systemic inflammation, and dialytic therapies, as well as factors that lead to the development of sarcopenia in older patients with CKD.

The prevalence of PEW significantly rises as the disease progresses, with rates increasing from <2% in CKD stages 1–2 to 11%–54% in CKD stages 3–5.¹¹ Presence or absence of PEW is one of the strongest predictors of poor clinical outcomes, including declining kidney function, hospitalization, and mortality among patients with CKD^{12–15} and has been associated with a two-fold increase in mortality risk in patients on maintenance hemodialysis.¹⁴ Careful nutritional management is essential in preventing the development of PEW and related morbidity and mortality in elderly patients with CKD. With the rapidly aging population and increasing burden of CKD, age- and CKD-related poor nutrition continues to be one of the main aspects of management in patients with kidney disease.

Etiology of PEW and Sarcopenia in Elderly Patients with CKD

A multitude of mechanisms are related to the development of PEW in advanced kidney disease (discussed in detail in refs. 16 and 17) (Figure 1). A common cause is decreased dietary intake due to anorexia affecting up to 40% of patients on maintenance dialysis.¹⁸ Patients with earlier CKD stages also spontaneously decrease their caloric intake as the disease progresses. Declining kidney function leads to metabolic alterations that result in suppression of appetite and changes in taste perception. Presence of systemic inflammation is also associated with anorexia in

chronic disease states, including CKD. Inflammation is a well-established promoter of muscle protein breakdown leading to sarcopenia.¹⁹ Among 328 patients with ESKD, the prevalence of high serum c-reactive protein and IL-6 was significantly higher among patients with PEW, defined as Subjective Global Assessment (SGA) score >1, independent of BMI. In multivariate analysis, presence of inflammation defined as c-reactive protein ≥ 10 mg/L was a significant predictor of PEW.²⁰ Similarly, in a study including 100 patients with CKD not on dialysis, high sensitivity c-reactive protein levels were inversely associated with appendicular lean mass adjusted for BMI while IL-4, an anti-inflammatory marker, was positively correlated with lower extremity lean mass.²¹ Metabolic acidosis in advanced kidney disease further aggravates muscle protein catabolism and is a factor that can be alleviated with oral bicarbonate supplementation. Maintenance dialysis is a major catabolic procedure that increases energy and protein requirements through nutrient loss from the hemodialysis membranes and increased inflammation from indwelling catheters and bioincompatible membranes. Various comorbidities seen in patients with CKD, including diabetes mellitus and gastrointestinal disturbances, may further reduce dietary nutrient intake relative to the needs.

Aging is similarly associated with loss of appetite and reduced food intake in the general population without CKD, termed anorexia of aging.²² Loss of smell and taste with increasing age negatively affects appetite and variety of the diet. Abnormalities in gastric motility and slower gastric emptying seen in the elderly lead to early satiety. Chronic low-grade inflammation, hormonal changes, and increased catabolism also contribute to poor nutritional status in older persons. Furthermore, nutritional intake in the elderly can be affected by many factors including problems with chewing, physical function impairment, cognitive dysfunction, acute and chronic diseases, low socioeconomic status, depression, and other social and environmental conditions.^{22–24}

Nutritional Screening and Assessments in Elderly Patients with CKD

Assessment of nutritional status is crucial for identifying poor nutritional intake and PEW and guiding nutritional

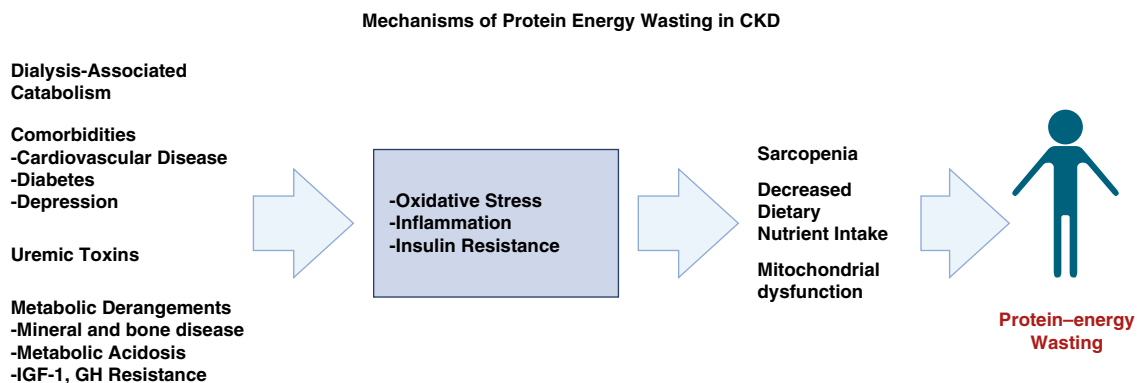


Figure 1. Mechanisms of protein–energy wasting in CKD. DEI, daily energy intake; DPI, daily protein intake; GH, growth hormone; IDPN, intradialytic parenteral nutrition; IGF-1, insulin-like growth factor type 1; ONS, oral nutritional supplementation; SGA, Subjective Global Assessment.

therapy. Various tools are available for assessing nutritional status and PEW with limited evidence to recommend one over another. The National Kidney Foundation Disease Outcomes Quality Initiative Clinical Practice Guideline for Nutrition in CKD: 2020 Update provides a comprehensive list of potential assessment methods, as summarized in Table 1. For the geriatric patients with CKD, a combination of methods readily available within the clinic can be used.

Routine screening for nutritional status should be conducted at least biannually in patients with CKD 3–5D. Factors that are commonly encountered in elderly patients and indirectly affect nutritional intake, such as medication use, knowledge, cognitive function, behavior, access to food, and depression, should be included in the assessment. On the basis of the recommendations of the National Kidney Foundation Disease Outcomes Quality Initiative 2020 guidelines, the preferred approach is to use comprehensive assessment tools, including SGA score or Malnutrition Inflammation Score (MIS). SGA score and MIS are simple and easily accessible methods that can be used in clinical settings and take approximately 10 minutes.²⁵ SGA includes information on nutrient intake; changes in weight; symptoms affecting oral intake; functional capacity; metabolic requirement; and physical examination findings, including loss of body fat, muscle mass, and presence of edema. Assessing nutritional status solely on the basis of biochemical markers, such as albumin, is not advised because these markers may be affected by factors other than nutrition, including fluid status, systemic inflammation, proteinuria, and residual kidney function. In addition, in this population, the presence of edema and volume expansion due to advancing CKD can mask the loss of nutritional reserves, making body weight an inaccurate measure of nutritional status. While routine nutritional screening with SGA or MIS biannually can be completed by any health care provider and requires minimal training, comprehensive assessment to guide intervention is preferentially provided by a registered dietitian or international equivalent in collaboration with a physician or other providers (nurse practitioners and physician assistants). This approach, referred to as medical

nutrition therapy, tailors the dietary intervention on the basis of nutritional assessment, comorbidities, and individual needs. Medical nutrition therapy includes the determination of barriers to adequate nutritional intake, offers nutritional education, and focuses on behavioral change.

A retrospective analysis conducted in the United States investigated the survival benefit of dietitian involvement before the onset of ESKD in older patients (aged 67 years and older). The study cohort consisted of 156,440 patients, with 88% receiving no dietitian care and only 3% receiving dietitian care for more than 1 year before initiating dialysis. Compared with patients with no dietitian care, patients with over 1 year of predialysis dietitian care had significantly higher serum albumin concentrations at initiation of maintenance dialysis. Among patients with the second tertile of propensity score for dietitian care, over 1 year of predialysis dietitian care was associated with lower risk of mortality over the follow-up period of 1 year. Although patients without dietitian care are more likely to also lack nephrologist care, which is a potential confounder, these results suggest that early nutritional management by a specialized team may improve survival in elderly patients with CKD.²⁶

Management of Nutrition in Elderly Patients with CKD

The initial management of nutrition in CKD focuses on preventive measures to minimize the loss of protein and energy stores. The recommended minimum dietary protein and caloric intakes for patients with stage 3–5 CKD not on dialysis are of 0.55–0.60 g/kg of body weight per day and 25–35 kcal/kg of body weight per day, respectively. This recommendation is supported by measurements of total energy expenditure in elderly patients (aged 60–86 years) with CKD stage 3–5 not on dialysis.²⁷

In specific cases, higher protein intake may be warranted. For patients with diabetes and CKD, a protein intake of 0.6–0.8 g/kg per day is recommended to achieve glycemic control. Patients on RRT, such as dialysis, may require a higher protein intake of 1.0–1.2 g/kg to maintain their

Table 1. Diagnostic tools for protein–energy wasting in elderly patients with CKD

Body composition	DEXA is considered the gold standard for the assessment of body composition and can be used in patients with CKD 1–5D. It should be noted that DEXA measurements are affected by volume status in patients on dialysis. Therefore, bioimpedance or MF-BIA should be preferred in patients on dialysis
Biochemical markers	For nutritional assessment, biochemical markers, including serum albumin, prealbumin or nPCR, should be coupled with other tools. Serum albumin can be used for predicting hospitalization and mortality in patients with CKD 5D
Handgrip strength	Repetitive handgrip strength can be used as an indicator of changes in functional status and protein energy status over time
Energy requirements	Given that indirect calorimetry, the gold standard for the estimation of resting energy expenditure, is not readily available in most clinical settings, disease-specific predictive energy equations can be used
Composite nutritional indices	The seven-point SGA score or MIS can be used for CKD 5D
Measurement of dietary intake	A 3-d food record can be used in patients with CKD 3–5D

Source: National Kidney Foundation Disease Outcomes Quality Initiative 2020. DEXA, dual-energy x-ray absorptiometry; MF-BIA, multifrequency bioelectrical impedance; MIS, Malnutrition Inflammation Score; nPCR, normalized protein catabolic rate; SGA, Subjective Global Assessment.

nutritional status.¹² Notably, these recommendations may vary depending on individual patient factors and should be tailored to each patient’s specific needs.

In cases where dietary intake is inadequate to prevent energy loss or evidence of PEW, nutritional supplementation is the mainstay of management. The indications for initiation of nutritional intervention and goals of therapy are demonstrated in Figure 2. Dietary supplementation should be given by oral route whenever possible (Figure 2). Oral supplementation leads to improvement in biochemical biomarkers, including serum albumin, prealbumin and transferrin concentrations, anthropometric measurements, and functional capacity, as quickly as within a month.¹⁷

If oral route is not feasible because of dysphagia or cognitive dysfunction, enteral tube feeding could be initiated in select patients. Enteral tube feeding can be administered through nasogastric tubes, percutaneous endoscopic gastroscopy, or jejunostomy tubes. There is no current evidence to support increased benefit from enteral feeding in elderly patients with CKD, and the decision for enteral feeding should be individualized considering each patient’s needs and barriers to oral feeding. When oral or enteral intake cannot be tolerated or is inadequate, parenteral nutrition may be necessary. Intradialytic parenteral nutrition (IDPN) has been shown to be safe and effective in patients on maintenance hemodialysis. While the benefits of oral and parenteral nutritional supplementation have been studied in various settings, there is limited research specifically investigating their effects in the geriatric CKD population. The available evidence on the

effects of nutritional supplementation in this population is summarized in Table 2. It is important to discuss enteral and parenteral feeding options, including intensity and duration, upfront with elderly patients or their surrogates because their goals in terms of quality of life could differ.

Oral Nutritional Supplementation

Multiple studies examined the effectiveness of oral nutritional supplementation (ONS) in the management for sarcopenia in the elderly population. In general, ONS has been shown to benefit elderly patients with evident frailty or malnutrition while no benefit has been proven in the general older population.⁴⁰ In a recent meta-analysis of 17 randomized clinical trials, ONS, given once to three times a day in liquid or solid form, was shown to significantly improve overall appetite, energy and protein intake, as well as body weight and BMI in the older population with malnutrition or an underlying disease.⁴¹ Another meta-analysis including 11 randomized clinical trials in elderly adults with frailty who were malnourished or at risk of malnutrition found little evidence on the effects of ONS energy intake, protein intake, and mobility.⁴² The mixed results in the current literature may be attributed to the various definitions of frailty and sarcopenia, as well as inconsistent use of assessment tools and modalities of ONS.⁴²

Studies specifically focusing on the elderly population with CKD are limited. Nevertheless, considering that the prevalence of CKD increases with age and the mean age of initiation of maintenance dialysis in CKD is reported to be

Nutritional Intervention in Elderly Patients with CKD

Indications:	Goals:
• DPI<1.2 or <0.7 g/kg/d for CKD5D and CKD 3-4, respectively • DEI <30 kcal/kg/d • Unintentional weight loss > 5% of ideal body weight over 3 months • Serum albumin <3.8 g/dL • Serum prealbumin <28 mg/dL • 7-point SGA score <6 • Loss of subcutaneous fat stores or muscle mass	• DPI > 1.2 and >0.7 g/kg/d for CKD 5D and CKD 3-4, respectively • DEI 30-35 kcal/kg/d • Serum albumin >4 g/dl • Serum prealbumin >30 mg/dL

Special Considerations:

- Start with oral nutritional supplementations (ONS) whenever possible. ONS can be taken at home or during hemodialysis sessions.
- In cases of no improvement, try increasing the quantity of ONS; consider switching to another ONS preparation on the basis of taste and texture preferences. If adequate oral intake cannot be reached, consider IDPN in patients on HD.
- Consider enteral tube feeding when swallowing is compromised.
- Consider parenteral nutrition when gastrointestinal tract is compromised.

Figure 2. Indications, goals, and special considerations for nutritional intervention in elderly patients with CKD.

Table 2. Studies assessing nutritional interventions in elderly patients with CKD

Intervention	Study	Sample Size	Time	CKD Stage	Mean Age, yr	Inclusion and Exclusion Criteria	Aim	Nutritional Intervention	Outcome
ONS	Hiroshige <i>et al.</i> ²⁸	44	12 mo	Hemodialysis	>70	Exclusion criteria included malnutrition due to factors unrelated to hemodialysis and inability to exercise	Improvement in plasma levels of BCAA and nutritional markers	Oral BCAA supplement for a total of 12 g/d (3.3 g valine, 5.7 g leucine, and 3 g isoleucine)	Significant increase in lean body mass and serum albumin
	Bolasco <i>et al.</i> ²⁹	30	3 mo	Hemodialysis	75.2±11.2	Hemodialysis for at least 6 mo, mean serum albumin of <3.5 g/dl, normalized protein nitrogen appearance <1.1 g/kg per day, and BMI >20 kg/m ²	Improvement in nutritional and inflammatory markers	4 g amino acids daily	Significant increase in serum albumin and total protein with amino acid supplementation
	Caglar <i>et al.</i> ³⁰	85	6 mo	Hemodialysis	62.3±15.4	Hemodialysis for at least 6 mo, mean serum albumin of ≤3.7 g/dl or mean serum prealbumin of <30 mg/dl	Improvement in nutritional markers	One Nepro (475 calories, 16.6 g protein, every hemodialysis session)	Significant increase in serum albumin, prealbumin, and SGA scores with intradialytic supplementation
	Kalantar-Zadeh <i>et al.</i> ³¹	41	4 wk	Hemodialysis	60.4±13.0	Hemodialysis patients with serum albumin <3.8 g/dl	Improvement in the serum albumin level	One Nepro (475 calories, 16.6 g protein) and one Oxepa (355 calories and 14.8 g protein) every hemodialysis session	Significant increase in the serum albumin level in patients with baseline albumin <3.8 g/dl
	Ewers <i>et al.</i> ³²	40	6 wk	Hemodialysis	64.6	Exclusion criteria included BMI >30 kg/m ² , serum CRP >10 mg/L accepted as an indication of infection	Improvement in dietary intake, blood lipid, and nutritional markers	Oral unsaturated fat supplement daily (430 kcal, 47 g fat, 26.5 g monounsaturated, and 3 g marine <i>n</i> -3 polyunsaturated fatty acids)	Significant increase in dry body weight, serum albumin, and CRP levels with unsaturated fat supplementation
IDPN	Hiroshige <i>et al.</i> ³³	28	12 mo	Hemodialysis	>70	Age >70 on hemodialysis	Improvement in nutritional markers	Parenteral nutrition formula (including glucose, essential amino acids, fat, 800 kcal) every hemodialysis session	Significant increase in anthropometric measurements, serum albumin, and transferrin
	Mortelmans <i>et al.</i> ³⁴	16	9 mo	Hemodialysis	65.5±9	Hemodialysis for at least 3 mo, weight loss by >6% over 1 yr or a prealbumin level <0.29 g/L	Improvement in nutritional markers	Parenteral nutrition formula (glucose, essential and nonessential amino acids, fat) every hemodialysis session	Significant increase in body weight and fat tissue mass
	Liu <i>et al.</i> ³⁵	32	9 mo	Hemodialysis	Control 71.80±9.51 Glucose 74.0±7.50 Amino acid 69.83±9.56	Hemodialysis for at least 3 mo, serum albumin <35 g/L, serum prealbumin or transferrin <200 mg/L or normalized protein decomposition/PCR <1.1 g/kg·d	Improvement in nutritional markers	Oral supplementation, oral supplementation plus high-concentration glucose solution and these two interventions plus 8.5% amino acids solution every hemodialysis session	Significantly higher serum albumin levels in the amino acid group compared with the glucose group in non-diabetic patients

Table 2. (Continued)

Intervention	Study	Sample Size	Time	CKD Stage	Mean Age, yr	Inclusion and Exclusion Criteria	Aim	Nutritional Intervention	Outcome
	Marsen <i>et al.</i> ³⁶	107	4 mo	Hemodialysis	IDPN 73.3±11.8 Control 75.0±8.48	Hemodialysis for at least 6 mo, SGA level B or C and two of the following criteria: albumin <35 g/L, prealbumin <250 mg/L or phase angle α <4.5° (BIA analysis)	Improvement in nutritional markers	Parenteral nutrition formula (glucose, amino acids, fat, vitamins and trace elements) every hemodialysis session	Significant increase in serum prealbumin in patients with PEW
	Kittiskulnam <i>et al.</i> ³⁷	38	3 mo	Hemodialysis	67.6±10.8	Hemodialysis for at least 3 mo, spontaneous dietary intake $\geq$ 20 kcal/kg per day and protein intake $\geq$ 0.8 g/kg per day, and presence of two of the following: albumin level $\leq$ 3.5 g/dl, prealbumin $\leq$ 30 mg/dl, SGA category B, or MIS $\geq$ 5 points	Improvement in nutritional markers	Parenteral nutrition formula (glucose, amino acids and fish oil-based lipid emulsion with omega-3 fatty acids, 1100 kcal) every hemodialysis session	Significant increase in body weight, serum albumin, spontaneous dietary intake and MIS with fish oil-based IDPN in patients with PEW intolerable to oral supplementation
	Cano <i>et al.</i> ³⁸	186	1 yr	Hemodialysis	Control 67.2±10.8 IDPN 68.8±9.9	Hemodialysis for at least 6 mo, and at least two of the following: BMI <20 kg/m ² , body weight loss within 6 mo >10%, serum albumin <35 g/L, and serum prealbumin <300 mg/L	All-cause mortality, hospitalization rate, Karnofsky performance score, and nutritional markers	IDPN plus oral supplements or oral supplements alone (IDPN regimen individualized to fulfil the difference between spontaneous and recommended intakes; oral supplementation 500 kcal/d and 25 g/d protein) every hemodialysis session	No additional nutritional or survival benefit of IDPN compared with oral supplementation alone
rhGH	Iglesias <i>et al.</i> ³⁹	17	4 wk	Hemodialysis and PD	63.9±3.1 in the rhGH group 58.3±5.6 in the control group	Peritoneal or hemodialysis for at least 6 mo, at least one of the following: weight loss >10% in the last 6 mo or >5% in the last month, reduction of >20% of MAMC and/or ideal body weight according to sex and age, serum albumin <4.0 g/dl, PCR <0.9 g/kg per day, and lymphocyte count <1000/ μ l	Improvement in anthropometric or laboratory nutritional markers and serum IGF-1	Recombinant human erythropoietin therapy (weekly mean dose, 102.8 $\pm$ 15.9 U/kg subcutaneously)	Significant increase in anthropometric measurements in the rhGH group

BCAA, branched-chain amino acid; BIA, bioelectrical impedance; BMI, body mass index; CRP, c-reactive protein; IDPN, intradialytic parenteral nutrition; IGF-1, insulin-like growth factor type 1; MAMC, mid-arm muscle circumference; MIS, malnutrition inflammation score; ONS, oral nutritional supplementation; PCR, protein catabolic rate; PD, peritoneal dialysis; rhGH, recombinant human growth hormone; SGA, Subjective Global Assessment.

60–73 years,^{43–45} the findings of most nutritional studies in CKD are applicable to the elderly population with CKD. Overall, available studies in older patients on maintenance hemodialysis indicate that oral supplementation is an effective strategy to improve nutritional status in this population, albeit the evidence is limited. Commercially available oral supplement formulations designed for patients with CKD typically contain a combination of amino acids, glucose, and fats. Renal-specific supplements provide high protein and energy content with minimal phosphate and potassium.⁴⁶ These supplements may be given at home or during dialysis sessions to increase compliance. The cost of commercially available oral supplement formulations with amino acids is approximately 3\$ per shake can.

Limited data from small pilot studies suggest that intradialytic ONS administered over 3–6 months improve certain nutritional markers, including serum albumin, prealbumin, and SGA scores.³⁰ The average age in these studies was around 60 years, and the energy content ranged from 475 calories to 830 calories per administration. In general, side-effect profiles were acceptable with gastrointestinal issues in three patients; no dropouts were reported.

The efficacy and safety of supplementation with specific essential nutrients have also been studied. A crossover study assessed the effects of unsaturated fat dietary supplements on nutritional and inflammatory markers in 40 patients on maintenance hemodialysis with a mean age of 65 years. Six weeks of supplementation increased the total energy intake (34 kcal/kg body weight per day) and resulted in significantly increased dry body weight and improved serum albumin and C-reactive protein levels. No change was observed in blood lipid concentrations despite the use of fat supplements. The adherence was worse than other studies in patients with CKD with seven of the 40 patients dropping out, five because of gastrointestinal side effects,³² indicating the issues with adherence to ONS. A pilot clinical trial including 30 patients on hemodialysis aged 75.2 ± 11.2 years assessed the efficacy of a special oral amino acid supplementation, including all essential amino acids plus tyrosine and cystine. Among the 15 patients randomized to receive 12 g of oral amino acid supplementation per day, significant elevations in serum albumin and total protein were seen over 3 months.²⁹ Oral supplementation of branched-chain amino acids for 6 months was also found to significantly increase lean body mass and serum albumin in patients with CKD older than 70 years.²⁸

IDPN

Parenteral infusion of nutrients during dialysis is a safe and convenient approach for patients who cannot tolerate oral or enteral administration of nutrients. Studies suggest that the combination of IDPN with oral nutritional supplements can improve markers of nutritional status, including dry body weight, skinfold measurements, and mid-arm muscle circumference, as well as serum albumin and prealbumin, in patients who cannot reach nutritional goals with oral supplements alone.^{28,36,37}

Earlier studies showed that IDPN can be safely used in the geriatric population. In a study including 78 elderly patients older than 70 years, IDPN was found to improve serum albumin and transferrin after 3 months and body

anthropometric measurements after 6 months of treatment. The mean age of the patients receiving IDPN was 81 ± 7 years, and no adverse events related to IDPN were reported over 1 year of IDPN.³³ Another study including 107 older patients (mean age was 73.3 ± 11.8 years) with underlying PEW reported that IDPN significantly increased prealbumin over 16 weeks while no significant change was observed in serum albumin or SGA scores. Seven patients in the IDPN group discontinued therapy because of adverse events, which included hyperglycemia, muscle cramps during hemodialysis, and gastrointestinal side effects.³⁶

Several studies compared the effects of different formulations of IDPN in elderly patients. Among 36 non-diabetic elderly hemodialysis patients, the group receiving amino acid-based IDPN had significantly higher albumin levels compared with the group receiving glucose-based IDPN after a treatment period of 9 months. There was no significant change in SGA scores compared with baseline in either group.³⁵

A randomized clinical trial performed in Thailand including 38 patients with PEW intolerable to ONS showed that fish oil-based IDPN for 3 months significantly improved body weight, serum albumin, spontaneous dietary intake, and MIS. The beneficial effects on serum albumin were maintained at 3 months after the cessation of treatment. The mean age was 67.6 ± 10.8 years, and two patients with manageable hyperglycemia were observed in the IDPN group.³⁷

While IDPN is valuable in the management of patients who cannot tolerate adequate oral supplementation, it is unlikely to provide additional nutritional benefit when sufficient oral supplementation can be maintained. A randomized trial including 186 patients with a mean age of 68.8 ± 9.9 years showed that adding IDPN to a regimen of oral supplementation that contained 500 kcal/d and 25 g/d protein did not offer nutritional or survival benefits in a follow-up period of 2 years. In both groups, body weight, serum albumin, and prealbumin increased after 1 year of oral supplementation without the additional effect of IDPN.³⁸

Adjunctive Therapies

Certain pharmacologic agents, with or without nutritional supplements, have been proposed to enhance anabolic activity or stimulate appetite in patients with CKD. These agents include recombinant human growth hormone (rhGH), nandrolone decanoate (ND), megestrol acetate, cyproheptadine, melatonin, thalidomide, and ghrelin. Most of these drugs have not been investigated in elderly patients with CKD, but their safety beyond short durations in this population is unclear.

Among these agents, rhGH has been shown to benefit nutritional status in various catabolic states, including patients on maintenance dialysis. In adults, rhGH enhances protein synthesis and reduces proteolysis in the muscle. Short-term rhGH administration in patients on hemodialysis improves net muscle protein balance and increases lean body mass while the effects in the elderly patients with CKD have not been studied in detail. In a pilot randomized clinical trial in patients with ESKD with a mean age of 63.9 ± 3.1 years, the addition of rhGH

to dietary prescriptions of 35 kcal/kg and 1 g protein/kg ideal body weight per day for 4 weeks significantly increased dry body weight and transferrin levels compared with dietary prescription alone.³⁹ The OPPORTUNITY trial assessed the efficacy of 24 months of rhGH treatment in reducing mortality in 695 patients with hypoalbuminemia on dialysis. The study was prematurely closed because of delayed recruitment issues. The interim analysis showed that while some cardiovascular risk factors, including body weight, body fat, serum inflammatory markers, and cholesterol levels, improved in the rhGH arm compared with placebo, there were no differences in all-cause and cardiovascular mortality, cardiovascular events, or nutritional markers.⁴⁷

Testosterone deficiency is a common endocrine disorder seen in up to 55% of male patients with advanced CKD.⁴⁸ Low serum levels of testosterone in CKD are associated with exaggerated sarcopenia and PEW, as well as increased mortality.^{49,50} According to the British Society for Sexual Medicine guidelines, testosterone therapy is indicated in men with symptoms of hypogonadism and total serum testosterone concentration <8 nmol/L or free testosterone <180 pmol/L.⁵¹ The aims of testosterone supplementation in CKD include improvement of quality of life, muscle mass, and functional status. In a recent study including 25 men with stage 3–4 CKD and testosterone deficiency, testosterone supplementation with 2% testosterone gel at 60 mg/d for 3 months was found to improve symptoms of hypogonadism as well as grip strength and sense of physical well-being. No significant change was seen in BMI.⁵² ND, a modified androgen analog of testosterone, was also efficacious in increasing lean body mass and muscle strength in patients on dialysis.^{53,54} Among 29 patients on maintenance hemodialysis, treatment with 100 mg ND for 6 months resulted in an increased lean body mass and quadriceps muscle cross-sectional area, measured by magnetic resonance imaging. The mean age was 56±13 years with a wide range of 26–88 years.⁵⁵ In a phase II dose-finding study, ND increased appendicular lean mass in a dose-responsive manner up to 200 mg/wk in men with stage 5 CKD. The mean age of the patients on the high dose was 53.6±7.0 years.⁵⁶ While testosterone supplementation may be an effective management to prevent sarcopenia, the efficacy in the elderly population remains unclear. The potential side effects, including volume overload, lipid disorders, increased risk of thrombosis, and alterations in liver function tests, should be monitored closely.

Sarcopenia and PEW are highly prevalent in elderly patients with moderate-to-advanced CKD and is associated with poor outcome. Multiple factors including reduced dietary caloric intake, metabolic and hormonal derangements related to advanced kidney disease and increasing age, as well as various barriers to nutrient intake including swallowing dysfunction, cognitive dysfunction, and other social determinants are involved in the development of these sarcopenia and PEW. Nutritional status should be assessed and screened routinely by health care providers and should guide the preventive and therapeutic interventions. Nutritional supplementation is generally effective in improving nutritional status and treating PEW and should be administered orally or enterally as indicated. Nutritional supplementations should be prescribed with considerations

to availability, tolerability, and elderly patients' specific needs and wishes.

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References

1. Sabharwal S, Wilson H, Reilly P, Gupta CM. Heterogeneity of the definition of elderly age in current orthopaedic research. *Springerplus*. 2015;4:516. doi:10.1186/s40064-015-1307-x
2. Kooman JP, Kotanko P, Schols AMWJ, Shiels PG, Stenvinkel P. Chronic kidney disease and premature ageing. *Nat Rev Nephrol*. 2014;10(12):732–742. doi:10.1038/nrneph.2014.185
3. Rosenberg IH. Sarcopenia: origins and clinical relevance. *J Nutr*. 1997;127(5 suppl):990S–991S. doi:10.1093/jn/127.5.990S
4. Kim JC, Kalantar-Zadeh K, Kopple JD. Frailty and protein-energy wasting in elderly patients with end stage kidney disease. *J Am Soc Nephrol*. 2013;24(3):337–351. doi:10.1681/ASN.2012010047
5. Cruz-Jentoft AJ, Baeyens JP, Bauer JM, et al. Sarcopenia: European consensus on definition and diagnosis: report of the European working group on sarcopenia in older people. *Age Ageing*. 2010;39(4):412–423. doi:10.1093/ageing/afq034
6. Fielding RA, Vellas B, Evans WJ, et al. Sarcopenia: an undiagnosed condition in older adults. Current consensus definition: prevalence, etiology, and consequences. International working group on sarcopenia. *J Am Med Dir Assoc*. 2011;12(4):249–256. doi:10.1016/j.jamda.2011.01.003
7. Shafiee G, Keshtkar A, Soltani A, Ahadi Z, Larijani B, Heshmat R. Prevalence of sarcopenia in the world: a systematic review and meta-analysis of general population studies. *J Diabetes Metab Disord*. 2017;16:21. doi:10.1186/s40200-017-0302-x
8. Lamarca F, Carrero JJ, Rodrigues JCD, Bigogno FG, Fetter RL, Avesani CM. Prevalence of sarcopenia in elderly maintenance hemodialysis patients: the impact of different diagnostic criteria. *J Nutr Health Aging*. 2014;18(7):710–717. doi:10.1007/s12603-014-0505-5
9. Giglio J, Kamimura MA, Lamarca F, Rodrigues J, Santin F, Avesani CM. Association of sarcopenia with nutritional parameters, quality of life, hospitalization, and mortality rates of elderly patients on hemodialysis. *J Ren Nutr*. 2018;28(3):197–207. doi:10.1053/j.jrn.2017.12.003
10. Fouque D, Kalantar-Zadeh K, Kopple J, et al. A proposed nomenclature and diagnostic criteria for protein-energy wasting in acute and chronic kidney disease. *Kidney Int*. 2008;73(4):391–398. doi:10.1038/sj.ki.5002585
11. Koppe L, Fouque D, Kalantar-Zadeh K. Kidney cachexia or protein-energy wasting in chronic kidney disease: facts and numbers. *J Cachexia Sarcopenia Muscle*. 2019;10(3):479–484. doi:10.1002/jcsm.12421
12. Ikizler TA, Cuppari L. The 2020 updated KDOQI clinical Practice guidelines for nutrition in chronic kidney disease. *Blood Purif*. 2021;50(4-5):667–671. doi:10.1159/000513698
13. Dai L, Mukai H, Lindholm B, et al. Clinical global assessment of nutritional status as predictor of mortality in chronic kidney

- disease patients. *PLoS One*. 2017;12(12):e0186659. doi: [10.1371/journal.pone.0186659](https://doi.org/10.1371/journal.pone.0186659)
14. de Mutsert R, Grootendorst DC, Boeschoten EW, et al. Subjective global assessment of nutritional status is strongly associated with mortality in chronic dialysis patients. *Am J Clin Nutr*. 2009;89(3):787–793. doi: [10.3945/ajcn.2008.26970](https://doi.org/10.3945/ajcn.2008.26970)
 15. Zhou Y, Hellberg M, Svensson P, Höglund P, Clyne N. Sarcopenia and relationships between muscle mass, measured glomerular filtration rate and physical function in patients with chronic kidney disease stages 3–5. *Nephrol Dial Transplant*. 2018;33(2):342–348. doi: [10.1093/ndt/gfw466](https://doi.org/10.1093/ndt/gfw466)
 16. Ikizler TA. Optimal nutrition in hemodialysis patients. *Adv Chronic Kidney Dis*. 2013;20(2):181–189. doi: [10.1053/j.ackd.2012.12.002](https://doi.org/10.1053/j.ackd.2012.12.002)
 17. Ikizler TA, Cano NJ, Franch H, et al. Prevention and treatment of protein energy wasting in chronic kidney disease patients: a consensus statement by the International Society of Renal Nutrition and Metabolism. *Kidney Int*. 2013;84(6):1096–1107. doi: [10.1038/ki.2013.147](https://doi.org/10.1038/ki.2013.147)
 18. Bossola M, Tazza L, Giungi S, Luciani G. Anorexia in hemodialysis patients: an update. *Kidney Int*. 2006;70(3):417–422. doi: [10.1038/sj.ki.5001572](https://doi.org/10.1038/sj.ki.5001572)
 19. Deger SM, Hung AM, Gamboa JL, et al. Systemic inflammation is associated with exaggerated skeletal muscle protein catabolism in maintenance hemodialysis patients. *JCI Insight*. 2017;2(22):e95185. doi: [10.1172/jci.insight.95185](https://doi.org/10.1172/jci.insight.95185)
 20. Honda H, Qureshi AR, Axelsson J, et al. Obese sarcopenia in patients with end-stage renal disease is associated with inflammation and increased mortality. *Am J Clin Nutr*. 2007;86(3):633–638. doi: [10.1093/ajcn/86.3.633](https://doi.org/10.1093/ajcn/86.3.633)
 21. Souza VA, Oliveira D, Barbosa SR, et al. Sarcopenia in patients with chronic kidney disease not yet on dialysis: analysis of the prevalence and associated factors. *PLoS One*. 2017;12(4):e0176230. doi: [10.1371/journal.pone.0176230](https://doi.org/10.1371/journal.pone.0176230)
 22. Landi F, Calvani R, Tosato M, et al. Anorexia of aging: risk factors, consequences, and potential treatments. *Nutrients*. 2016;8(2):69. doi: [10.3390/nu8020069](https://doi.org/10.3390/nu8020069)
 23. Roubenoff R. Sarcopenia and its implications for the elderly. *Eur J Clin Nutr*. 2000;54(suppl 3):S40–S47. doi: [10.1038/sj.ejcn.1601024](https://doi.org/10.1038/sj.ejcn.1601024)
 24. Dhillion RJ, Hasni S. Pathogenesis and management of sarcopenia. *Clin Geriatr Med*. 2017;33(1):17–26. doi: [10.1016/j.cger.2016.08.002](https://doi.org/10.1016/j.cger.2016.08.002)
 25. Rambod M, Bross R, Zitterkoph J, et al. Association of Malnutrition-Inflammation Score with quality of life and mortality in hemodialysis patients: a 5-year prospective cohort study. *Am J Kidney Dis*. 2009;53(2):298–309. doi: [10.1053/j.ajkd.2008.09.018](https://doi.org/10.1053/j.ajkd.2008.09.018)
 26. Slinin Y, Guo H, Gilbertson DT, et al. Prehemodialysis care by dietitians and first-year mortality after initiation of hemodialysis. *Am J Kidney Dis*. 2011;58(4):583–590. doi: [10.1053/j.ajkd.2011.03.032](https://doi.org/10.1053/j.ajkd.2011.03.032)
 27. D'Alessandro C, Giannese D, Avino M, Cupisti A. Energy requirement for elderly CKD patients. *Nutrients*. 2021;13(10):3396. doi: [10.3390/nu13103396](https://doi.org/10.3390/nu13103396)
 28. Hiroshige K, Sonta T, Suda T, Kanegae K, Ohtani A. Oral supplementation of branched-chain amino acid improves nutritional status in elderly patients on chronic haemodialysis. *Nephrol Dial Transplant*. 2001;16(9):1856–1862. doi: [10.1093/ndt/16.9.1856](https://doi.org/10.1093/ndt/16.9.1856)
 29. Bolasco P, Caria S, Cupisti A, Secci R, Saverio Dioguardi F. A novel amino acids oral supplementation in hemodialysis patients: a pilot study. *Ren Fail*. 2011;33(1):1–5. doi: [10.3109/0886022X.2010.536289](https://doi.org/10.3109/0886022X.2010.536289)
 30. Caglar K, Fedje L, Dimmitt R, Hakim RM, Shyr Y, Ikizler TA. Therapeutic effects of oral nutritional supplementation during hemodialysis. *Kidney Int*. 2002;62(3):1054–1059. doi: [10.1046/j.1523-1755.2002.00530.x](https://doi.org/10.1046/j.1523-1755.2002.00530.x)
 31. Kalantar-Zadeh K, Braglia A, Chow J, et al. An anti-inflammatory and antioxidant nutritional supplement for hypoalbuminemic hemodialysis patients: a pilot/feasibility study. *J Ren Nutr*. 2005;15(3):318–331. doi: [10.1016/j.jrn.2005.04.004](https://doi.org/10.1016/j.jrn.2005.04.004)
 32. Ewers B, Riserus U, Marckmann P. Effects of unsaturated fat dietary supplements on blood lipids, and on markers of malnutrition and inflammation in hemodialysis patients. *J Ren Nutr*. 2009;19(5):401–411. doi: [10.1053/j.jrn.2009.04.006](https://doi.org/10.1053/j.jrn.2009.04.006)
 33. Hiroshige K, Iwamoto M, Kabashima N, Mutoh Y, Yuo K, Ohtani A. Prolonged use of intradialysis parenteral nutrition in elderly malnourished chronic haemodialysis patients. *Nephrol Dial Transplant*. 1998;13(8):2081–2087. doi: [10.1093/ndt/13.8.2081](https://doi.org/10.1093/ndt/13.8.2081)
 34. Mortelmans AK, Duym P, Vandenbroucke J, et al. Intradialytic parenteral nutrition in malnourished hemodialysis patients: a prospective long-term study. *JPEN J Parenter Enteral Nutr*. 1999;23(2):90–95. doi: [10.1177/014860719902300290](https://doi.org/10.1177/014860719902300290)
 35. Liu Y, Xiao X, Qin DP, et al. Comparison of intradialytic parenteral nutrition with glucose or amino acid mixtures in maintenance hemodialysis patients. *Nutrients*. 2016;8(6):220. doi: [10.3390/nu8060220](https://doi.org/10.3390/nu8060220)
 36. Marsen TA, Beer J, Mann H.; German IDPN-Trial group. Intradialytic parenteral nutrition in maintenance hemodialysis patients suffering from protein-energy wasting. Results of a multicenter, open, prospective, randomized trial. *Clin Nutr*. 2017;36(1):107–117. doi: [10.1016/j.clnu.2015.11.016](https://doi.org/10.1016/j.clnu.2015.11.016)
 37. Kittiskulnam P, Banjongjit A, Metta K, et al. The beneficial effects of intradialytic parenteral nutrition in hemodialysis patients with protein energy wasting: a prospective randomized controlled trial. *Sci Rep*. 2022;12(1):4529. doi: [10.1038/s41598-022-08726-8](https://doi.org/10.1038/s41598-022-08726-8)
 38. Cano NJ, Fouque D, Roth H, et al. Intradialytic parenteral nutrition does not improve survival in malnourished hemodialysis patients: a 2-year multicenter, prospective, randomized study. *J Am Soc Nephrol*. 2007;18(9):2583–2591. doi: [10.1681/ASN.2007020184](https://doi.org/10.1681/ASN.2007020184)
 39. Iglesias P, Díez JJ, Fernández-Reyes MJ, et al. Recombinant human growth hormone therapy in malnourished dialysis patients: a randomized controlled study. *Am J Kidney Dis*. 1998;32(3):454–463. doi: [10.1053/ajkd.1998.v32.pm9740162](https://doi.org/10.1053/ajkd.1998.v32.pm9740162)
 40. Gammack JK, Sanford AM. Caloric supplements for the elderly. *Curr Opin Clin Nutr Metab Care*. 2015;18(1):32–36. doi: [10.1097/MCO.0000000000000125](https://doi.org/10.1097/MCO.0000000000000125)
 41. Li M, Zhao S, Wu S, Yang X, Feng H. Effectiveness of oral nutritional supplements on older people with anorexia: a systematic review and meta-analysis of randomized controlled trials. *Nutrients*. 2021;13(3):835. doi: [10.3390/nu13030835](https://doi.org/10.3390/nu13030835)
 42. Thomson KH, Rice S, Arisa O, et al. Effectiveness and cost-effectiveness of oral nutritional supplements in frail older people who are malnourished or at risk of malnutrition: a systematic review and meta-analysis. *Lancet Healthy Longev*. 2022;3(10):e654–e666. doi: [10.1016/S2666-7568\(22\)00171-4](https://doi.org/10.1016/S2666-7568(22)00171-4)
 43. Bonenkamp AA, Hoekstra T, Hemmelder MH, et al. Trends in home dialysis use differ among age categories in past two decades: a Dutch registry study. *Eur J Clin Invest*. 2022;52(1):e13656. doi: [10.1111/eci.13656](https://doi.org/10.1111/eci.13656)
 44. Fu EL, Evans M, Carrero JJ, et al. Timing of dialysis initiation to reduce mortality and cardiovascular events in advanced chronic kidney disease: nationwide cohort study. *BMJ*. 2021;375:e066306. doi: [10.1136/bmj-2021-066306](https://doi.org/10.1136/bmj-2021-066306)
 45. Cooper BA, Branley P, Bulfone L, et al. A randomized, controlled trial of early versus late initiation of dialysis. *New Engl J Med*. 2010;363(7):609–619. doi: [10.1056/NEJMoa1000552](https://doi.org/10.1056/NEJMoa1000552)
 46. Mah JY, Choy SW, Roberts MA, et al. Oral protein-based supplements for people with chronic kidney disease requiring dialysis. *Cochrane Database Syst Rev*. 2017;2017(3):CD012616. doi: [10.1002/14651858.CD012616](https://doi.org/10.1002/14651858.CD012616)
 47. Kopple JD, Cheung AK, Christiansen JS, et al. OPPORTUNITY: a large-scale randomized clinical trial of growth hormone in hemodialysis patients. *Nephrol Dial Transplant*. 2011;26(12):4095–4103. doi: [10.1093/ndt/gfr363](https://doi.org/10.1093/ndt/gfr363)
 48. Albaaj F, Sivalingham M, Haynes P, et al. Prevalence of hypogonadism in male patients with renal failure. *Postgrad Med J*. 2006;82(972):693–696. doi: [10.1136/pgmj.2006.045963](https://doi.org/10.1136/pgmj.2006.045963)
 49. Romejko K, Rymarz A, Sadownik H, Niemczyk S. Testosterone deficiency as one of the major endocrine disorders in chronic kidney disease. *Nutrients*. 2022;14(16):3438. doi: [10.3390/nu14163438](https://doi.org/10.3390/nu14163438)
 50. Skiba R, Matyjek A, Syryło T, Niemczyk S, Rymarz A. Advanced chronic kidney disease is a strong predictor of hypogonadism and is associated with decreased lean tissue mass. *Int J Nephrol Renovasc Dis*. 2020;13:319–327. doi: [10.2147/IJNRD.S275554](https://doi.org/10.2147/IJNRD.S275554)

51. Hackett G, Kirby M, Edwards D, et al. British Society for Sexual medicine guidelines on adult testosterone deficiency, with statements for UK practice. *J Sex Med*. 2017;14(12):1504–1523. doi:[10.1016/j.jsxm.2017.10.067](https://doi.org/10.1016/j.jsxm.2017.10.067)
52. Yeo JK, Koo HS, Yu J, Park MG. Effects of testosterone treatment on quality of life in patients with chronic kidney disease. *Am J Mens Health*. 2020;14(3):1557988320917258. doi:[10.1177/1557988320917258](https://doi.org/10.1177/1557988320917258)
53. Barton Pai A, Chretien C, Lau AH. The effects of nandrolone decanoate on nutritional parameters in hemodialysis patients. *Clin Nephrol*. 2002;58(1):38–46. doi:[10.5414/cnp.58038](https://doi.org/10.5414/cnp.58038)
54. Johansen KL, Mulligan K, Schambelan M. Anabolic effects of nandrolone decanoate in patients receiving dialysis: a randomized controlled trial. *JAMA*. 1999;281(14):1275–1281. doi:[10.1001/jama.281.14.1275](https://doi.org/10.1001/jama.281.14.1275)
55. Johansen KL, Painter PL, Sakkas GK, Gordon P, Doyle J, Shubert T. Effects of resistance exercise training and nandrolone decanoate on body composition and muscle function among patients who receive hemodialysis: a randomized, controlled trial. *J Am Soc Nephrol*. 2006;17(8):2307–2314. doi:[10.1681/ASN.2006010034](https://doi.org/10.1681/ASN.2006010034)
56. Macdonald JH, Marcora SM, Jibani MM, Kumwenda MJ, Ahmed W, Lemmey AB. Nandrolone decanoate as anabolic therapy in chronic kidney disease: a randomized phase II dose-finding study. *Nephron Clin Pract*. 2007;106(3):c125–c135. doi:[10.1159/000103000](https://doi.org/10.1159/000103000)

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